PASSWORD CAPTURING USING WIRESHARK

Objective: Demonstrate how packet captures can expose information transmitted through an unencrypted HTTP connection, using Wireshark in an authorized laboratory environment.

Important: Perform packet capture only on networks, devices, and applications for which you have permission. HTTPS/TLS is designed to prevent credentials from appearing as readable HTTP form data.

1. Introduction

Wireshark is a network protocol analyzer capable of displaying packets captured from a network interface. For unencrypted protocols such as HTTP, captured application data may be visible in packet details. This report demonstrates the analysis workflow shown in the supplied screenshots: start a capture, generate web traffic, filter HTTP packets, distinguish GET and POST requests, and inspect the submitted form data.

Workflow demonstrated

1. Start Wireshark and capture traffic **2.** Open the demonstration login page **3.** Filter HTTP traffic **4.** Check GET requests **5.** Check POST requests **6.** Expand the HTML form URL encoded section to inspect submitted fields.

Useful display filters shown in the screenshots

```
http
http.request.method == "GET"
http.request.method == "POST"
```

The screenshots below are the primary evidence for each stage; they are intentionally retained as screenshots rather than replaced with generic diagrams.

Step 1 — Start Wireshark packet capture

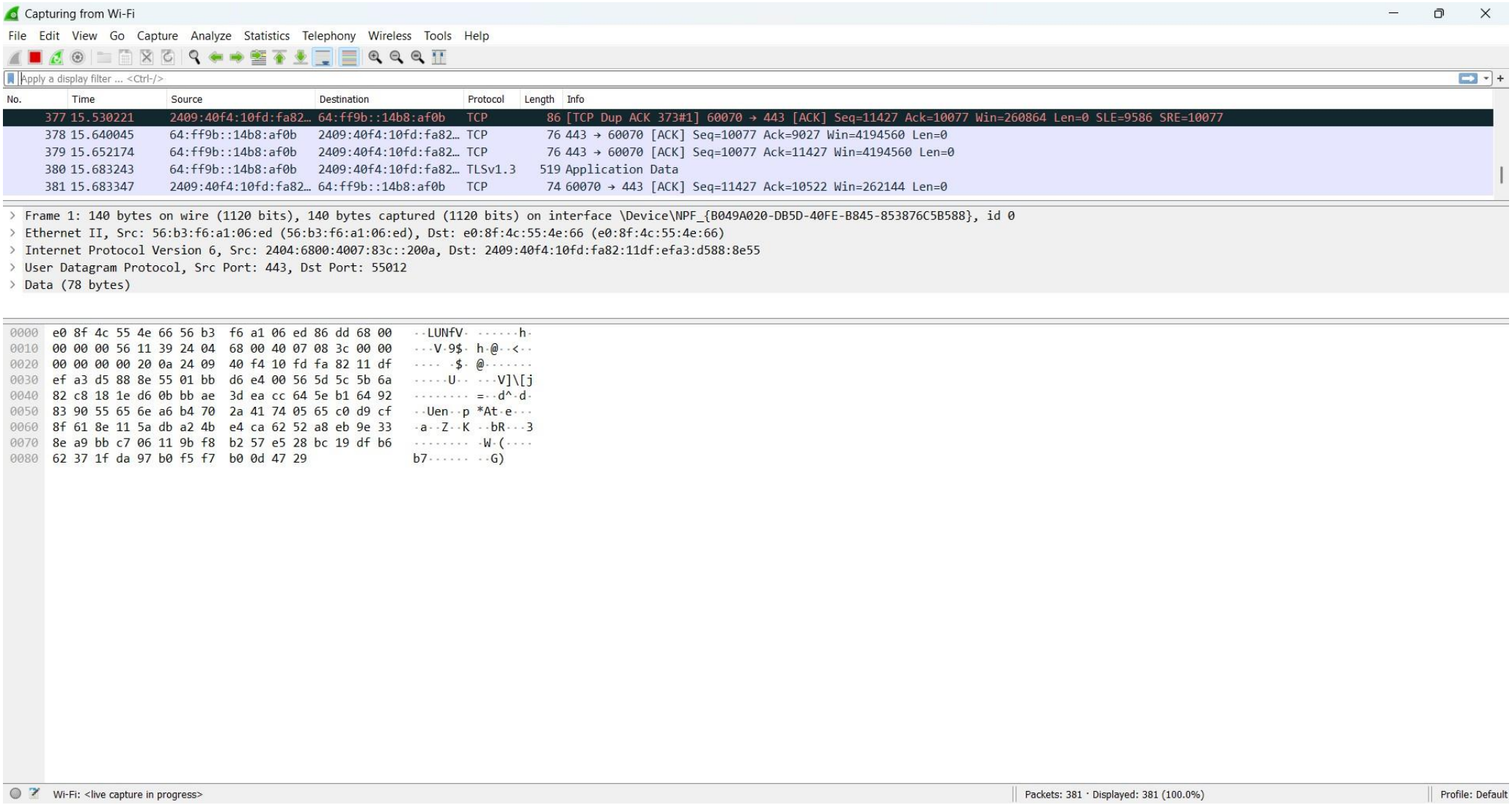

Figure 1. Step 1 — Start Wireshark packet capture.

The capture is running on the Wi-Fi interface. Packets are being collected before and during the web activity.

Step 2 — Login page and submitted credentials

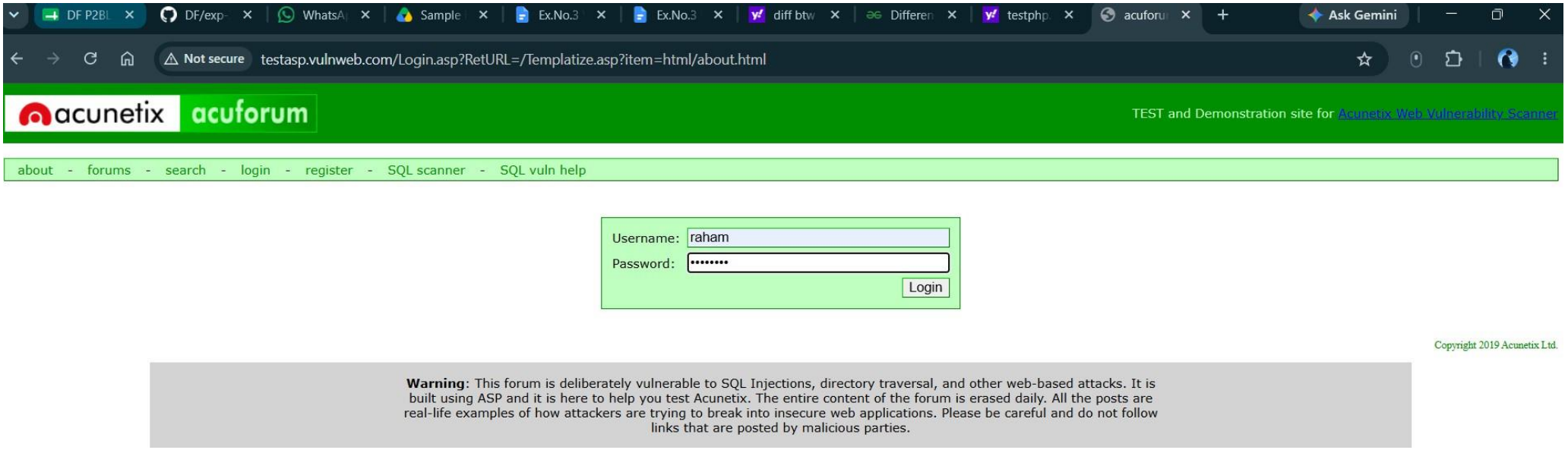

Figure 2. Step 2 — Login page and submitted credentials.

The demonstration website displays a login form.

Step 3/4 — Filtering captured traffic for HTTP

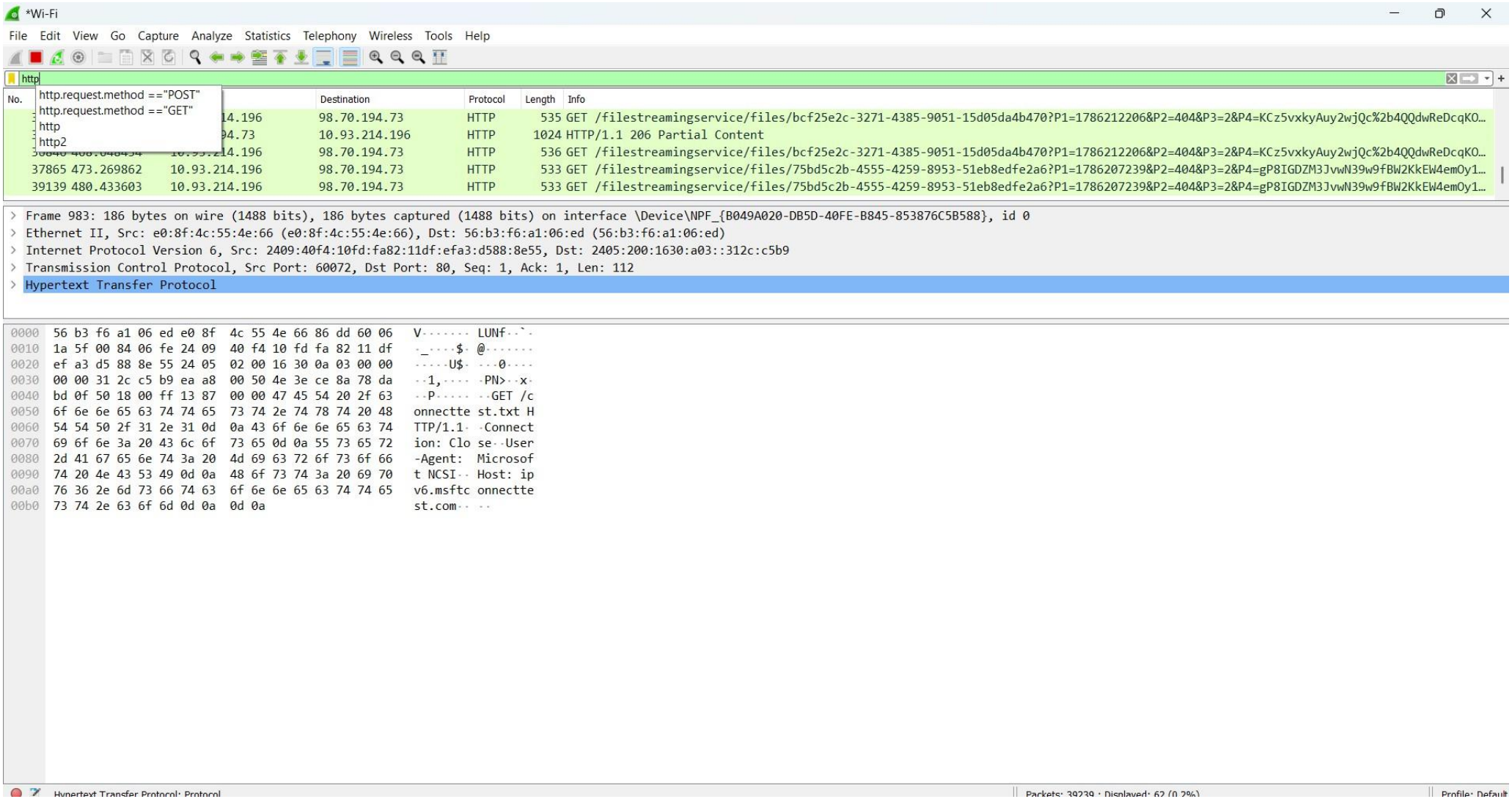

Figure 3. Step 3/4 — Filtering captured traffic for HTTP.

The display filter `http` is used to narrow the packet list to HTTP traffic.

Step 5/6 — Inspecting HTTP GET requests

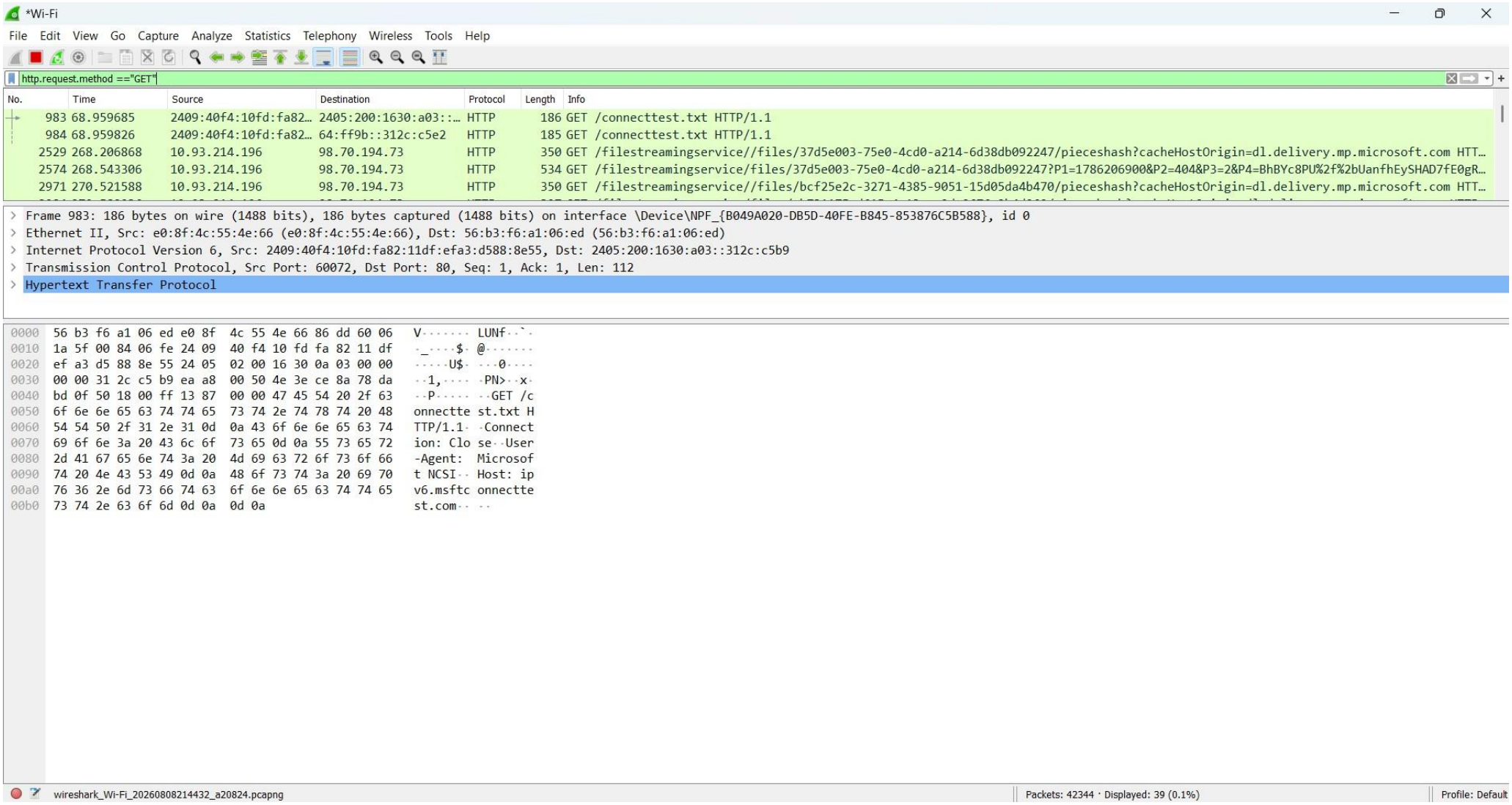

Figure 4. Step 5/6 — Inspecting HTTP GET requests.

The GET filter shows HTTP GET requests. A GET request does not necessarily contain the login form submission; the POST request is checked next.

Step 7 — Inspecting the HTTP POST form data

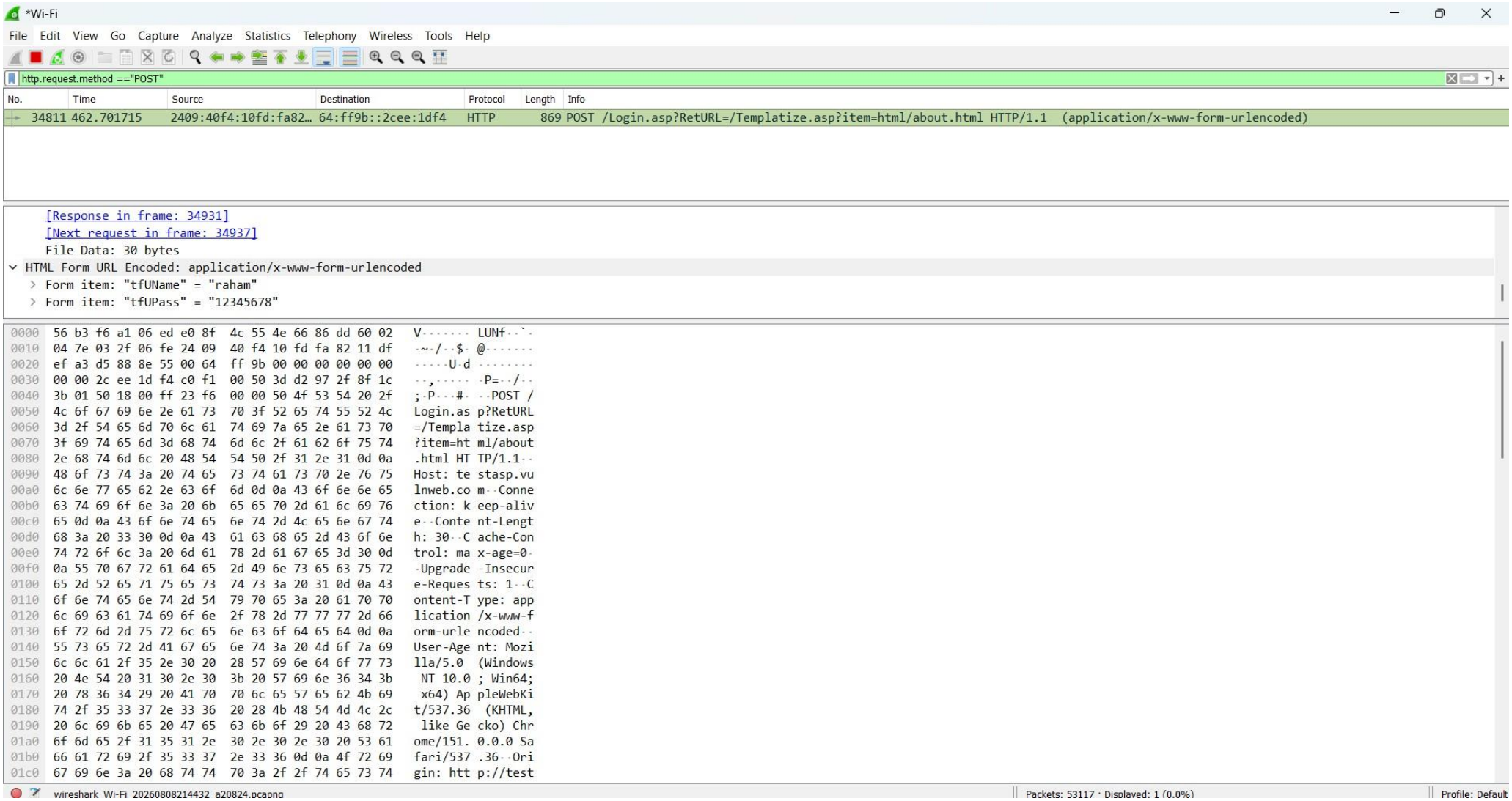

Figure 5. Step 7 — Inspecting the HTTP POST form data.

The POST packet is selected and the HTML Form URL Encoded section exposes the submitted form fields. This is possible because the demonstration traffic is plain HTTP rather than encrypted HTTPS.

2. Result and Conclusion

The supplied capture demonstrates that Wireshark can reveal application-layer information when that information is transmitted over an unencrypted protocol such as HTTP. By filtering the capture with HTTP, GET, and POST display filters, the relevant login transaction can be located and its form-encoded fields inspected.

Security takeaway: Credentials should never be transmitted over plain HTTP. Web applications should enforce HTTPS/TLS, use secure session management, and protect sensitive information in transit.

Evidence included

All five supplied screenshots are included in their original visual form …